Steven R. Dunbar
Department of Mathematics
203 Avery Hall
University of Nebraska-Lincoln
Lincoln, NE 68588-0130
http://www.math.unl.edu
Voice: 402-472-3731
Fax: 402-472-8466

Topics in Probability Theory and Stochastic Processes
Steven R. Dunbar

A Markov Chain Model of Baseball

Rating

Mathematically Mature: may contain mathematics beyond calculus with proofs.

Section Starter Question

What is the baseball statistic known as OPS? How is this statistic used?

Key Concepts

1. The goal is to model a half-inning of a baseball game as a Markov chain to find the expected number of runs a player will score as a means of evaluating and comparing players.

2. The expected number of runs a player will score uses the standard partition of the transition probability matrix over the transient and absorbing states.

Vocabulary

1. The key new statistic is how many runs will a team score if it uses this player for every plate appearance, called **Markov runs**.

2. The **On-base plus slugging** statistic or OPS is the sum of a player's slugging average and his on-base average.

Mathematical Ideas

This section assumes familiarity with the rules and terminology of American baseball and some of the statistics associated with baseball players. The goal is to model a half-inning of a baseball game as a Markov chain to find the expected number of runs a player will score as a means of evaluating and comparing players.

The Markov chain will be the sequence of situations arising in a half-inning of a baseball game. In this Markov chain model of baseball each step will be a plate appearance. A plate appearance is similar to an at-bat, but includes walks, hit batters, and sacrifices that recorded baseball does not regard as an official at-bat. In this baseball model, the states are the runners on base together with a count of the present outs. Each plate appearance has eight possibilities for the runners on base:

- the bases can be empty;
- there can be one runner on first, second, or third base;
- there can be two runners on base in three different ways; or
- the bases can be loaded.

The number of outs can be 0, 1, or 2 to make twenty-four states. All situations with three outs are the same, and this situation is the final 25th absorbing state. Once the game reaches this state, the half-inning ends and the Markov chain is stopped. This simplified model takes no account of how many players are on base or where players are located when the third out is made. Table 1 below gives the labeling of the 24 possible runners and outs states. Figure 1 is a diagram of the states along with a few sample transitions. The model tracks runs scored separately, runs are not part of the states.

Denote the probability of moving from one state i to state j by $p_{i,j}$. Many transitions are impossible, such as a transition to a state with fewer outs, and hence have probabilities equal to zero. Using the numbering above, $p_{1,4}$ is the probability of going from no outs and none on to a runner on first with no outs, which in turn is equal to the probability of a single plus the probability of a walk. In real baseball, there are other possibilities for this transition such

Runners:	None	1st	2nd	3rd	1&2	1&3	2&3	1,2,&3
Outs								
0	1	4	7	10	13	16	19	22
1	2	5	8	11	14	17	20	23
2	3	6	9	12	15	18	21	24
3	25							

Table 1: States of the Markov chain model of a half inning of baseball

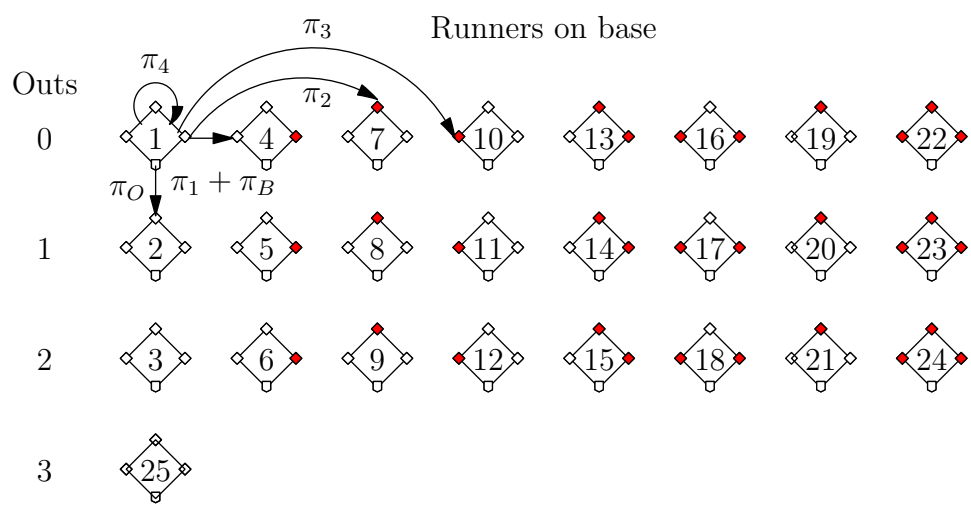

Figure 1: The state diagram for the Markov chain model of a half-inning of baseball, with only transitions from state 1 indicated as an example.

as a wild pitch, but this simplified model ignores them. In certain situations different actions by the batter produce the same effect. As can be seen in Figure 1, if no one is on base, then a single and a walk have the same effect. The probability of each event adds to give the state transition probability.

For this model imagine that the player bats randomly with plate outcomes distributed according to his statistical profile. Plate appearances are independent. In other words, a player's chance of getting a specific kind of hit or making a specific kind of out is independent of how many runners are on and the number of outs. Based on this profile, compute the transition probabilities for each pair of states, obtaining a Markov chain. The transition probabilities could come from player statistics from the previous year, the current year, or the career, but must use the same profile each time. The transition probability matrix P, whose entries are $p_{i,j}$ governs the evolution of the half inning. By the properties of matrix multiplication, P^2 is the transition matrix for sequences of two plays or batters, P^3 is the transition matrix for sequences of three batters, and so on, for a string of identical batters. This assumption is useful for computing *average* performance which is the goal here, but not for specific game situations.

Encoding the half inning of the game in the transition matrix like this requires simplifying assumptions.

1. The Markov property assumption means that we don't care how we arrived at a particular situation. For example, if there is a runner on first and no outs, the Markov chain model is not concerned about whether there was a walk or a single.

2. More specifically, the Markov property assumption deliberately eliminates the possibility of a *clutch hitter*. Such a player allegedly hits better when the situation matters more, but numerical studies indicate that the concept is an illusion.

3. This model ignores situations such as stolen bases or pick-offs that could occur during the plate appearance, changing the state without action from the player in his plate appearance.

4. The model ignores double plays, sacrifice bunts, and sacrifice flies.

5. The model ignores runs scored while incurring the third out.

6. An important simplifying assumption is that the matrix is time independent, and each plate appearance is an independent event. In real baseball, as pitchers and batters face each other multiple times, each adjusts to the other, changing the probabilities. Fatigue may also be a factor as a game progresses.

7. Another related assumption is that the batter's outcomes do not depend on the pitcher and what the pitcher does is not dependent on the hitter. The probabilities remain constant, regardless of the current match-up and previous encounters between pitcher and hitter.

8. A related question is whether players get "hot" or "cold", the model ignores this possibility.

9. Already mentioned above is the assumption that a single advances a runner by two bases. For example, in states 4, 5, or 6 with runner on first, a single by the hitter moves the base runner to third base, leaving the chain in states 16, 17, or 18 respectively. Also, assume that a double advances each runner two bases, so that in states 4, 5, or 6 with runner on first, a double by the hitter moves the base runner to third base, leaving the chain in states 19, 20, or 21. In contrast, in [1] Cover and Keilers explicitly make the assumption that "All singles and doubles are assumed to be long. That is, a single advances a baserunner two bases, and a double scores a runner from first base." On the other hand, Pankin [3] is not explicit about this assumption, but the sample calculations of R imply that more detailed base running plays have non-zero probability. For instance, Pankin implies the possibility that bases loaded with one out (state 23) can change to state 5 or 8 meaning that a runner on first, as well as the two runners on second and third, can score on a single or a double. Likewise Pankin implies that a change from state 23 to 23 is possible which would be a single allowing one run from third, advancing the runners on first and second one base. This is not assumed here.

Each assumption can be relaxed at the cost of more states and a larger transition probability matrix, or using more detailed player statistics or both. The intent here is to show how to model baseball as a Markov chain, in the process creating a new player evaluation statistic.

	AB	S	D	T	HR	BB	AVE	SA	OB	OPS
X	500	100	25	5	30	100	.320	.570	.433	1.003
MT	470	63	27	2	45	110	,291	.438	.645	1.083
CB	558	86	34	3	47	95	.305	.629	.406	1.035

Table 2: Statistics for representative players.

A particular batter H determines a twenty-five by twenty-five matrix M_H from his statistical profile. For convenience, write the player's profile as follows, deviating slightly from the standard listing. Here AB denotes at-bats, S denotes singles, D denotes doubles, T denotes triples, HR denotes home runs, BB denotes walks plus hit batters, AVE denotes batting average $(S + D + T + HR)/AB$, SA denotes slugging average $(1 \cdot S + 2 \cdot D + 3 \cdot T + 4 \cdot HR)/AB$, a measure of the batting productivity of a player, OB denotes on-base average $(BB + S + D + T + HR)/(AB + BB)$, and $OPS = OB + SA$ measuring the ability of a player both to get on base and to hit for power, two important offensive skills. Note that the last four elements depend on the first six, and hence are not needed. For this model, the number of plate appearances is $AB + BB$, ignoring sacrifice flies as in the official score-keeping. In this simplified scenario, all outs are equivalent, there is no practical difference between a strikeout and a pop-out. As a result, values of OB and OPS here differ slightly from the official records. These statistics give the complete profile of H for the model. In other words, we know the probabilities that H makes each kind of hit or out, draws a walk, gets hit by a pitch, and so on.

For example, Table 2 gives a fictional line for a hypothetical star player X. Table 2 also includes 2019 season profiles for Mike Trout (Los Angeles Angels, American League MVP for 2019) and Cody Bellinger (Los Angeles Dodgers, National League MVP for 2019). Player X has 500 at-bats but 600 plate appearances. In a given plate appearance player X has a $\frac{1}{6}$ chance to hit a single, a $\frac{1}{6}$ chance to reach first base by a walk or hit batter, a $\frac{1}{120}$ chance to hit a triple, and so on. In this way, from the 6 basic player statistics, compute a probability distribution $(\pi_O, \pi_B, \pi_1, \pi_2, \pi_3\pi_4)$, where π_O is the probability of an out, π_B is the probability of a base on balls, and π_i is the probability of an i-base hit.

A state can change in several ways, for instance state 16 with runners on first and third with no outs can change into states 17, 22, 16 itself while

scoring a run with a single, 19, 10 or 1. A convenient way to build the transition probability matrix is to compose it from 6 simpler matrices with entries either 0 or π_x, corresponding to the 6 probabilities in the hitter's distribution. For example, for π_O, the probability of an out, the number of outs increases by one and the base runners remain the same, which makes a regularly shaped transition matrix P_0, where states with an index divisible by 3 transition to the absorbing state 25 of three outs. All other states advance by 1 index with probability π_O. Taking P_3 as another example, a triple scores all runners already on base, leaves the hitter on third base and the numbers of outs stays the same. Thus in P_3 each state transitions to state 10, 11 or 12 according to the existing number of outs. The other 4 matrices are similar, but the advance of states is slightly more complicated because of the assumption that runners on base advance by 2 bases on a hit. Finally, the full probability transition matrix is the sum of the 6 elementary transition matrices: $P = P_O + P_B + P_1 + P_2 + P_3 + P_4$. Set $P_{25,25} = 1$ to account for the absorbing end state. The transition probability matrices are too large to display here, but the scripts below build the matrices given the vales for $(\pi_O, \pi_B, \pi_1, \pi_2, \pi_3\pi_4)$.

Keeping track of runs scored will be an important outcome of the model. The key new statistic is how many runs will a team score by using this player for every plate appearance. This new statistic is called **Markov runs**. Cover and Keilers [1] call this the *Offensive Earned-Run Average* or OREA but that term seems to be no longer used. This statistic, normalized using nine innings, for all players each day of the season is available at USA Today, Sagarin ratings. Given the player's statistical profile, the player bats according to the Markov chain model until making three outs. In principle the test is run thousands of times, and from it the model determines the average number of runs scored per nine innings. However, simple results from Markov chain theory calculate the statistics without needing thousands of simulations. From Sagarin ratings, a team consisting of (nine clones of) Mike Trout would average 8.72 runs per game.

Let R be the 25×1 column vector containing the expected or average runs scored after each state on one play only. Denoting the elements of R by

R_1, R_2, ..., R_{25} some example calculations are:

$$\begin{aligned}
R_1 &= p_{1,1}(\text{ based on a home run}) \\
R_2 &= p_{2,2}(\text{ based on a home run}) \\
R_6 &= p_{6,12} + 2p_{6,3}(\text{ based on triple or a home run}) \\
R_{17} &= p_{17,17} + p_{17,20} + 2p_{17,11} + 3p_{17,1} \\
R_{23} &= 2p_{23,17} + 2p_{23,20} + 3p_{23,11} + 4p_{23,2} \\
R_{25} &= 0.
\end{aligned}$$

The vector of runs calculated here will be componentwise slightly less than the run vector for Cover and Keilers because advances on singles and doubles are less than they assume. Likewise, the vector of runs calculated here will be componentwise less than the run vector for Pankin because some state transitions he considers are ignored here.

The key output of the Markov chain baseball model is the computation of the expected runs in the rest of the inning after any runners and outs state. Let E be the 25×1 column vector containing these values. Then,

$$E = R + PR + P^2R + P^3R + \cdots .$$

This equation says that the expected runs after any state is the sum of the expected runs after one plate appearance, the expected runs after two plate appearances, and so on theoretically forever. Of course, almost surely (in the probability sense!) the half-inning will end sometime in the 25th absorption state of three outs. Because the vector of runs calculated here is less than the other run vectors, the expected run vector, or Markov runs will be slightly less. Since the 25th state with 3 outs is the only absorbing state and the other 24 states are transient, the 25×25 probability transition matrix partitions into the block matrix

$$P = \begin{pmatrix} Q & Q' \\ 0 & 1 \end{pmatrix}$$

where Q is 24×24, and Q' is 24×1. Since no runs are scored from the 25th absorbing state, just consider

$$E' = R' + QR' + Q^2R' + Q^3R' + \cdots$$

where R' and E' are the 24×1 subvectors of E and R corresponding to the first 24 entries. By the standard matrix theory, this is equal to

$$E' = (I_{24} - Q)^{-1}R'.$$

The expected runs vector E' has useful interpretations. For example, if P includes all events for a league, then E' contains the average expected runs scored in the rest of the inning from each runners-and-outs state. With enough statistics, similar values for teams, a team's home and road games, or for an individual player can be calculated. In the last case, we obtain an estimate of run scoring if that player batted all the time. Also, by restricting the transitions to those not influenced by strategies such as stolen bases or sacrifice bunts, we obtain expected run values to compare with games using those strategies.

Note that E'_1 is the expected runs after the no runners, no out state in which all innings begin. Thus the **Markov runs** $9E'_1$ is the expected number of runs per 9 half-innings, or per game. This computation is especially interesting when applied to an individual player's statistics. It provides another way of estimating how many runs per game a player would score by batting all the time. As noted above, some baseball statisticians report this number for players, but much more common is the **On-base plus slugging** statistic or OPS. For OPS, simply add the player's slugging average to his on-base average. This simply computed player statistic empirically correlates fairly well with the expected number of runs scored using the Markov chain model. More details are in the article by D'Angelo,[2]

The elementary model here assumes a series of identical batters, of course this does not happen in real games. This has been the major criticism of the applicability of Markov models. But emphasizing again, it is possible to have a Markov model with different transition matrices for each batter. Then the assumption of stationarity is dropped and the probability transition matrix changes with each batter according to that batter's statistics. Instead of powers of one transition matrix P, use products of the matrices for each batter: P_1P_2, $P_1P_2P_3$, etc. Also, the expected runs on the next play column vector R has to be modified for each batter. With such changes, the expected runs generalizes to

$$E = R_1 + P_1R_2 + P_1P_2R_3 + P_1P_2P_3R_4 + \cdots .$$

An additional potential complication is that it may be necessary to repeat the calculation for several sequences with different first batters and then weight the results by the probability of specific batters beginning the sequence. The model can be expanded in a number of ways and the utility is limited only by the amount of data available.

Section Ending Answer

Sources

This section is adapted from the articles [1], [2], and [3]. Each of those articles has many further references. The online article The Markov Chain Model of Baseball has the same basic information as this article, but in a condensed form.

Algorithms, Scripts, Simulations

Algorithm

Scripts

```

pi_0 <- 333/596
pi_B <- 126/596
pi_1 <- 63/596
pi_2 <- 27/596
pi_3 <- 2/596
pi_4 <- 45/596

P0 <- matrix(0, nrow=25, ncol=25)
PB <- matrix(0, nrow=25, ncol=25)
P1 <- matrix(0, nrow=25, ncol=25)
P2 <- matrix(0, nrow=25, ncol=25)
```

```
P3 <- matrix(0, nrow=25, ncol=25)
P4 <- matrix(0, nrow=25, ncol=25)

##  State          1   2   3   4   5   6   7   8  9   10  11  12  13  14  15  16  17  18  19  20  21  22  23  24
transitionOut <- c(2,  3, 25,  5,  6, 25,  8,  9, 25, 11, 12, 25, 14, 15, 25, 17, 18, 25, 20, 21, 25, 23, 24, 25)
transitionBB  <- c(4,  5,  6, 13, 14, 15, 13, 14, 15, 16, 17, 18, 22, 23, 24, 22, 23, 24, 22, 23, 24, 22, 23, 24)
transition1B  <- c(4,  5,  6, 16, 17, 18,  4,  5,  6,  4,  5,  6, 16, 17, 18, 16, 17, 18, 19, 20, 21, 16, 17, 18)
transition2B  <- c(7,  8,  9, 19, 20, 21,  7,  8,  9,  7,  8,  9, 19, 20, 21, 19, 20, 21,  7,  8,  9, 19, 20, 21)
transition3B  <- c(10, 11, 12, 10, 11, 12, 10, 11, 12, 10, 11, 12, 10, 11, 12, 10, 11, 12, 10, 11, 12, 10, 11, 12)
transitionHR  <- c(1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3)

for (i in 1:24) {
    P0[i, transitionOut[i] ] <- pi_0
    PB[i, transitionBB[i]  ] <- pi_B
    P1[i, transition1B[i]  ] <- pi_1
    P2[i, transition2B[i]  ] <- pi_2
    P3[i, transition3B[i]  ] <- pi_3
    P4[i, transitionHR[i]  ] <- pi_4
}

P <- P0 + PB + P1 + P2 + P3 + P4
P[25, 25] = 1

M <- matrix(0, nrow=25, ncol=5)
M[22:24,   1] <- 1
M[7:18,  2:3] <- 1
M[19:24, 2:3] <- 2
M[4:12,    4] <- 1
M[13:21,   4] <- 2
M[22:24,   4] <- 3
M[1:3,     5] <- 1
M[4:12,    5] <- 2
M[13:21,   5] <- 3
M[22:24,   5] <- 4

trout <- c(pi_B, pi_1, pi_2, pi_3, pi_4)
runsTrout <- M %*% trout

```

```
Q <-  P[1:24, 1:24]
cat( solve(diag(24) - Q, runsTrout[1:24]) )

library("markovchain")
nStates <- 25
baseballStates <- as.character(c(1:nStates))
mcBaseball <- new("markovchain", states = baseballStates,
    byrow=TRUE, transitionMatrix = P, name="mcTrout")
cat( expectedRewards(mcBaseball, 30, runsTrout) )
```

Problems to Work for Understanding

1: Steven Strasburg of the Washington Nationals was the 2019 World Series MVP. Discuss why this model may not useful for assessing Strasburg as a baseball player.

2: Use this model to calculate the Markov runs for Ted Williams and Babe Ruth. Does this settle the debate about who is the greatest hitter in baseball history?

3: Consider a player K who each time at bat either hits a home run with probability q or strikes out with probability $1 - q$. Write out the non-zero probability transitions in the matrix M_K.

Reading Suggestion:

References

[1] Thomas M. Cover and Carroll W. Keilers. An offensive earned-run average for baseball. *Operations Research*, 25(5):729–740, Sep.-Oct. 1977.

[2] John P. D'Angelo. Baseball and markov chains: Power hitting and power series. *Notices of the American Mathematical Society*, 57(4):490–495, April 2010.

[3] Mark D. Pankin. Markov chain models:theoretical background. Accessed December 19, 2019.

Outside Readings and Links:

1.

2.

3.

4.

Solutions

1: In 2019 Strasburg was a pitcher for the Washington Nationals in the National League, as he has been since 2010. As a pitcher, his hitting skills and run production are not emphasized, it is his pitching that makes him a valuable player. Furthermore, as a pitcher, he appears in fewer games with even fewer plate appearances. That means he has a smaller base of hitting

	AB	S	D	T	HR	BB	AVE	SA	OB	OPS
Ruth	8399	1517	506	136	714	2062	0.342	0.690	0.474	1.164
Williams	7706	1534	525	71	521	2021	0.344	0.634	0.482	1.116

Table 3: Statistics for Babe Ruth and Ted Williams.

statistics for creating the matrix P so P may not fully reflect his hitting skills.

2: The lifetime career statistics for Babe Ruth (1914–1935) and Ted Williams (1939–1942, 1946–1960) are in Table 3. (Values derived from Baseball-Reference.) Then

$$\pi_{BR} = (0.412, 0.246, 0.181, 0.060, 0.016, 0.085)$$

and

$$\pi_{TW} = (0.394, 0.262, 0.199, 0.068, 0.009, 0.068).$$

Using these probabilities in the script for Markov runs, the value for Babe Ruth is 2.83 and the value for Ted Williams is 3.05. The two values are close but Williams has the slightly greater value.

No statistic can resolve the comparison between these two great players. Each played in different baseball eras, and each had particular strengths. Besides, what is to be the basis for comparison, their lifetime statistics used here, or the best career year for each? If the latter, how to choose the best career year? Note that Williams lost 3 (possibly best) years of his career to military service in World War II. It is interesting to compute the Markov runs for each, but the statistic won't answer this opinion question.

3: In the matrix M_K, all entries are 0 except for the following:

- For each i, $i = 1, \ldots, 25$ and $j = i \mod 3$, $(M_K)_{ij} = q$,
- For each i, $i = 1, \ldots, 25$ with $i \neq 0 \mod 3$, and $j = i+1$ $(M_K)_{ij} = 1-q$
- For each i, $i = 1, \ldots, 25$ with $i = 0 \mod 3$ and $j = 25$ $(M_K)_{ij} = 1-q$.

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Steve Dunbar's Home Page, `http://www.math.unl.edu/~sdunbar1`

Email to Steve Dunbar, `sdunbar1 at unl dot edu`

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